

PROFESSIONAL SUMMARY:

Senior Backend Python Software Engineer with 7+ years of experience designing, developing and maintaining enterprise-grade web applications and APIs using Python, Django and modern backend frameworks. Experienced in building secure, high-performance and scalable solutions for simulation, financial, manufacturing and consulting domains. Adept in RESTful API development, asynchronous programming, containerization and virtualization technologies like QEMU and Docker. Proficient in relational database design, performance tuning, and integration with distributed systems. Collaborator translating complex business requirements into technical implementations within Agile environments. Experience on Functional / Non-Functional requirements, Proof of Concepts (POCs), prototypes, Detailed Project Reports (DPRs), High-Level Design (HLD) and Low-Level Design (LLD) documentation, project scope, objectives, methodology, outcomes. Comprehensive understanding of the data science ecosystem, encompassing machine learning, Deep Learning, Computer Vision, Natural Language Processing, GANs, GenAI, RAG, BERT, LLMs models, MLOps and AIOps.

College Foundation of North Carolina

Senior Software Engineer

Raleigh, NC.

Mar 2024 - Present.

Responsibilities:

- Designed and implemented scalable backend components for data center simulation tools using Python, Django and RESTful APIs.
- Collaborated with product managers and architects to translate business requirements into secure, high-performance backend solutions.
- Developed asynchronous data processing pipelines to optimize simulation workloads and reduce processing latency.
- Containerized applications with Docker and Kubernetes for consistent deployment across multiple environments.
- Integrated relational database solutions using PostgreSQL, optimizing query performance for large-scale datasets.
- Conducted code reviews, mentoring junior developers on best practices for maintainable and clean code.
- Debugged complex virtualization workflows leveraging QEMU-based environments to support simulation testing.
- Implemented unit and integration tests to ensure production readiness and reliability.
- Utilized Git-based workflows within Agile sprints to deliver incremental, high-quality features on schedule.

Environment: Python, Django, REST API, PostgreSQL, Docker, Kubernetes, QEMU, Git, Agile, Asynchronous Programming.

Flexible Metal

Software Engineer

Atlanta, GA.

March 2023 - Feb2024.

Responsibilities:

- Built and maintained backend APIs for manufacturing process automation using Python and Django.
- Designed secure, high-throughput data exchange between simulation systems and manufacturing dashboards.
- Asynchronous programming patterns used to improve responsiveness in equipment monitoring applications.
- Integrated containerization strategies with Docker to ensure smooth deployment to staging and production.
- Developed SQL queries and schema designs for PostgreSQL databases to manage complex product data.

Vinay Srikakulapu **Senior Software Engineer**

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KEY SKILLS:

Python, Django, RESTful API Development, Asynchronous Programming, PostgreSQL, Docker, Kubernetes, QEMU, Linux Networking, API Security, Agile Methodologies, Git, CI/CD, Unit & Integration Testing, Performance Optimization, System Design, Data Structures and Algorithms.

TECHNICAL SKILLS:

- Programming Languages: Python, JavaScript, HTML, CSS.
- Frameworks & Tools: Django, FastAPI, Flask, Angular (integration).
- API Development: RESTful APIs, Asynchronous Programming, API Security.
- Databases: PostgreSQL, MySQL, SQLite.
- Virtualization: QEMU, Docker, Kubernetes.
- Networking: Linux Networking, SDN Concepts.
- Version Control & CI/CD: Git, GitLab CI/CD, GitHub Actions.
- Testing: Unit Testing, Integration Testing, PyTest.
- Methodologies: Agile (Scrum/Kanban), TDD.
- OS Platforms: Linux, Windows.
- AI Hardware Acceleration: GPUs (NVIDIA A100, V100, T4), TPUs (v2, v3), CUDA, cuDNN.

EDUCATION:

Masters in Information Systems,
Indiana Wesleyan University 2024.

- Partnered with cross-functional teams to troubleshoot networking and virtualization issues in production.
- Performed debugging and optimization of backend services for latency-sensitive workloads.
- Participated in agile ceremonies, contributing to sprint planning and backlog grooming for continuous delivery.
- Conducted peer reviews to maintain high coding standards and adherence to architectural guidelines.

Environment: Python, Django, REST API, PostgreSQL, Docker, Agile, Git, Asynchronous Programming, Networking Concepts.

Fidelity Investments

Role: Software Engineer

Durham, NC.

Sep 2020 - Nov 2022.

Responsibilities:

- Developed backend services supporting financial modeling applications with secure REST API endpoints in Django.
- Implemented data virtualization solutions using Docker and QEMU to replicate trading environments for testing.
- Designed scalable database schemas and optimized SQL queries to handle millions of financial transactions daily.
- Implemented asynchronous data handling to improve processing efficiency for portfolio analytics.
- Collaborated with architecture teams to design solutions meeting compliance, security, and performance requirements.
- Created automated unit and integration test suites to ensure code reliability across deployments.
- Debugged network-level and API-level issues in production to ensure minimal downtime.
- Participated in code reviews and pair programming sessions to maintain coding consistency across teams.
- Delivered features incrementally within Agile sprints using Git-based CI/CD workflows.

Environment: Python, Django, REST API, PostgreSQL, Docker, QEMU, Git, Agile, Asynchronous Programming, Financial Systems.

Thoughtworks

Software Engineer

India.

Jun 2018 - Aug 2020.

Responsibilities:

- Designed and delivered backend solutions for enterprise web applications using Python, Django, and REST architecture.
- Created asynchronous job queues to handle large-scale data ingestion and processing workloads.
- Implemented containerized deployments with Docker, enabling faster development-to-production cycles.
- Integrated secure authentication and authorization workflows for sensitive client data.
- Collaborated with front-end teams to align API contracts with Angular-based user interfaces.
- Optimized PostgreSQL database performance through indexing, query tuning and schema improvements.
- Debugged and resolved complex networking issues impacting service connectivity in distributed environments.
- Backend code reviews and adherence to clean code and design principles.
- Delivered projects within Agile frameworks, leveraging Git for version control and CI/CD automation.

Environment: Python, Django, REST API, PostgreSQL, Docker, Git, Agile, Asynchronous Programming, Networking Concepts, Angular Integration.

Domain: Data Center Simulation, Computer Hardware Manufacturing, Financial Services, Manufacturing Automation, IT Consulting.

PROJECTS:

Data Center Digital Twin Simulation Platform

- Developed a Python/Django-based platform to simulate and visualize virtual data centers.
- Integrated QEMU, Docker and Kubernetes for scalable multi-node environments.
- Built RESTful APIs and async scheduling for provisioning, monitoring and high-load traffic simulation.

Tech Stack: Python, Django, PostgreSQL, QEMU, Docker, Kubernetes, REST APIs, Async Programming.

API-Driven Manufacturing Process Optimizer

- Developed backend APIs and async pipelines for real-time IoT manufacturing telemetry.
- Optimized database schemas for large datasets and used Docker for consistent deployment.
- Collaborated with front-end teams to integrate Angular dashboards with backend services.

Tech Stack: Python, Django, PostgreSQL, Docker, REST APIs, WebSockets, Angular.

Networking Protocol Debugger Tool

- Developed a Linux CLI tool to debug packet flows in simulated SDNs.
- Integrated with QEMU and Docker lab environments.
- Created parsing logic to detect packet drops, latency spikes, and misconfigurations.

Tech Stack: Python, Scapy, QEMU, Docker, Linux Networking.